

FULL STACK APPLICATION DEVELOPMENT

SKILL – 3: Working with HQL - Sorting, Pagination & Aggregates

Aim

To perform advanced data retrieval operations using Hibernate Query Language (HQL) such as sorting, pagination, filtering, grouping, and aggregate functions.

Prerequisites

- Basic knowledge of Hibernate
- Basic idea of HQL syntax

Software Requirements

- Java 8
- MySQL
- Eclipse IDE
- Hibernate 5.6
- Maven

Task 1: Insert 5–8 Product Records

HQL / Hibernate Code:

```
session.save(product);
```

Output Screenshot:

id	description	name	price	quantity
3	Gaming Laptop	Laptop	2000000	12
4	Wireless Mouse	Mouse	1200	40
5	Gaming Laptop	Laptop	75000	10
6	Wireless Mouse	Mouse	1200	40
7	gaming	laptop	68000	7
8	normal	dell	78000	1
9	gaming	asus	98000	1
10	gaming	hp victus	10000	1
11	electronics	laptop	55000	5

Task 2: Sort Products by Price (Ascending)

from Product order by price asc

Output Screenshot:

```
1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit

Hibernate: select product0_id as id1_0_, product0_description as descript2_0_, product0_name as name3_0_, product0_price as price4_0_ from product order by price asc
Mouse 1200.0
Mouse 1200.0
hp victus 10000.0
laptop 55000.0
laptop 60000.0
laptop 75000.0
dell 78000.0
asus 98000.0
Laptop 2000000.0
```

Task 3: Sort Products by Price (Descending)

from Product order by price desc

Output Screenshot:

```
1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit

Hibernate: select product0_id as id1_0_, product0_description as descript2_0_, product0_name as name3_0_, product0_price as price4_0_ from product order by price desc
Laptop 2000000.0
asus 98000.0
dell 78000.0
laptop 75000.0
laptop 60000.0
laptop 55000.0
hp victus 10000.0
Mouse 1200.0
Mouse 1200.0
```

Task 4: Sort by Quantity (Highest First)

from Product order by quantity desc

Output Screenshot:

```
1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit

4
Hibernate: select product0_.id as id1_0_, product0_.description as descript2_0_, product0_.name as name3_0_, product0_.price as price4_0_
Mouse 40
Mouse 40
Laptop 12
Laptop 10
laptop 7
laptop 5
dell 1
asus 1
hp victus 1
```

Task 5: Pagination

```
Query q = session.createQuery("from Product"); q.setFirstResult(0);
q.setMaxResults(3);
```

Output Screenshot:

```
1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit

5
First 3 Products
Hibernate: select product0_.id as id1_0_, product0_.description as descript2_0_, product0_.name as name3_0_, product0_.price as price4_0_
com.klu.model.Product@31973858
com.klu.model.Product@67b04dcd
com.klu.model.Product@738d37fc
Next 3 Products:
Hibernate: select product0_.id as id1_0_, product0_.description as descript2_0_, product0_.name as name3_0_, product0_.price as price4_0_
com.klu.model.Product@1775c4e7
com.klu.model.Product@47829d6d
com.klu.model.Product@2f677247
```

Task 6: Aggregate Functions

```
select count(*) from Product
select min(price) from Product
select max(price) from Product
```

Output Screenshot:

```

1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit
11
Hibernate: select count(*) as col_0_0_ from product product0_
Hibernate: select count(*) as col_0_0_ from product product0_ where product0_.quantity>0
Hibernate: select product0_.description as col_0_0_, count(*) as col_1_0_ from product product0_ group by product0_.description
Hibernate: select min(product0_.price) as col_0_0_ from product product0_
Hibernate: select max(product0_.price) as col_0_0_ from product product0_
Total Products: 9
Quantity > 0: 9
Gaming Laptop 2
Wireless Mouse 2
gaming 3
normal 1
electronics 1
Min Price: 1200.0
Max Price: 200000.0

```

Task 7: Group By Description

select description, count(*) from Product group by description

Output Screenshot:

```

1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit
11
Hibernate: select product0_.description as col_0_0_, count(*) as col_1_0_ from product product0_ group by product0_.description
Gaming Laptop 2
Wireless Mouse 2
gaming 3
normal 1
electronics 1

```

Task 8: Filter Products Using WHERE

from Product where price between :min and :max

Output Screenshot:

```

1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit
11
Min price: 10000
Max price: 100000
Hibernate: select product0_.id as id1_0_, product0_.description as descript2_0_, product0_.name as name3_0_, product0_.price as price4_
com.klu.model.Product@7fda2001
com.klu.model.Product@11c713b7
com.klu.model.Product@2251b3bc
com.klu.model.Product@3f0e503c
com.klu.model.Product@3c232051

```

Task 9: LIKE Operations

```
name like 'A%'
name like '%p'
name like '%ou%'
length(name)=5
```

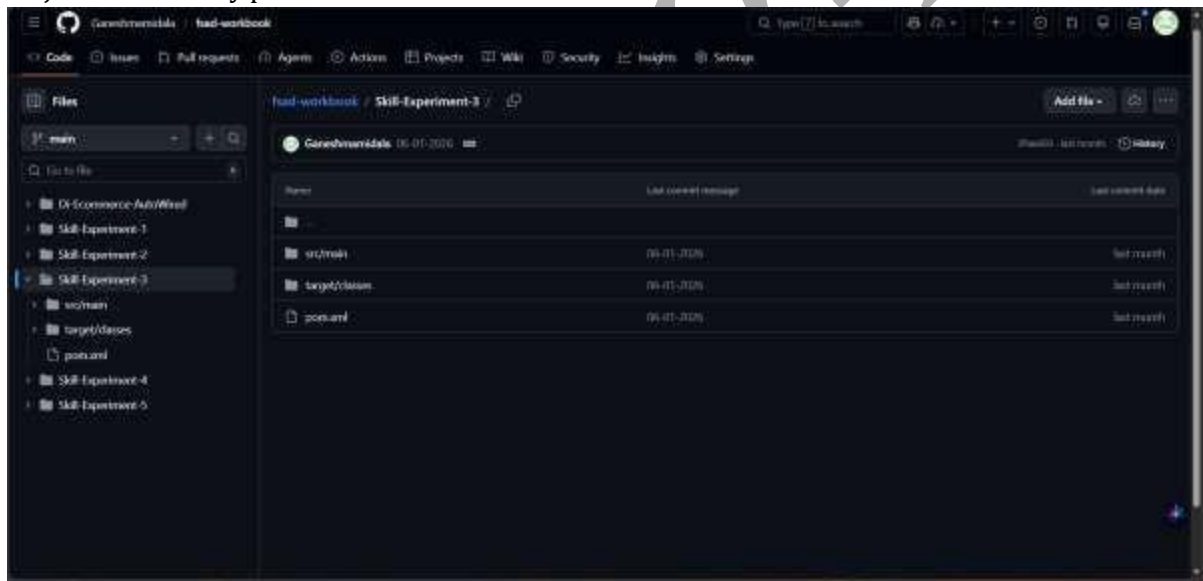
Output Screenshot:

```
1 Insert Product
2 Sort Price Ascending
3 Sort Price Descending
4 Sort Quantity Highest First
5 Pagination
6 Aggregate Functions
7 Group By Description
8 Filter Price Range
9 LIKE Operations
10 Exit

5
1 Starts With
2 Ends With
3 Contains
4 Exact Length
```

Task 10: GitHub Repository

Project successfully pushed to GitHub.



Result

All HQL operations including sorting, pagination, filtering, grouping and aggregate functions were successfully implemented using Hibernate.

GitHub Link : <https://github.com/Anil1843/fsad-skill>

Codes:

MainApp:

```
package com.klu.mainapp;
```

```
import com.klu.model.Product;

import org.hibernate.*;

import org.hibernate.cfg.Configuration;
```

```
import java.util.List;

import java.util.Scanner;
```

```
public class MainApp {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        SessionFactory sf = new Configuration().configure().buildSessionFactory();
```

```
        Session session = sf.openSession();
```

```
        Transaction tx = session.beginTransaction();
```

```
        System.out.println("1 Insert Product");
```

```
        System.out.println("2 Sort Price Ascending");
```

```
        System.out.println("3 Sort Price Descending");
```

```
        System.out.println("4 Sort Quantity Highest First");
```

```
        System.out.println("5 Pagination");
```

```
        System.out.println("6 Aggregate Functions");
```

```
        System.out.println("7 Group By Description");
```

```
        System.out.println("8 Filter Price Range");
```

```
        System.out.println("9 LIKE Operations");
```

```
        System.out.println("10 Exit");
```

```
        int ch = sc.nextInt();
```

```
switch (ch) {
```

```
    case 1:
```

```
        System.out.print("How many products: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 0; i < n; i++) {
```

```
            Product p = new Product();
```

```
            System.out.print("Name: ");
```

```
            p.setName(sc.next());
```

```
            System.out.print("Description: ");
```

```
            p.setDescription(sc.next());
```

```
            System.out.print("Price: ");
```

```
            p.setPrice(sc.nextDouble());
```

```
            System.out.print("Quantity: ");
```

```
            p.setQuantity(sc.nextInt());
```

```
            session.save(p);
```

```
        }
```

```
        tx.commit();
```

```
        System.out.println("Products inserted"); break;
```

```
    case 2:
```

```
        List<Product> asc = session.createQuery("from Product order by price asc",
```

```
        Product.class).list(); asc.forEach(p ->
```

```
            System.out.println(p.getName() + " " + p.getPrice()));
```

break;

case 3:

```
List<Product> desc = session.createQuery("from Product order by price desc",  
Product.class).list(); desc.forEach(p ->  
    System.out.println(p.getName() + " " + p.getPrice()));
```

break;

case 4:

```
List<Product> qty =  
    session.createQuery("from Product order by quantity desc",  
Product.class).list();  
qty.forEach(p ->  
    System.out.println(p.getName() + " " + p.getQuantity()));
```

break;

case 5:

```
Query q = session.createQuery("from Product");  
q.setFirstResult(0);  
q.setMaxResults(3);  
System.out.println("First 3 Products");  
q.list().forEach(System.out::println);  
q.setFirstResult(3);  
q.setMaxResults(3);  
System.out.println("Next 3 Products");  
q.list().forEach(System.out::println);
```

break;

case 6:

Long total =


```
(Long) session.createQuery("select count(*) from Product").uniqueResult();
```

```
Long available =
```

```
(Long) session.createQuery("select count(*) from Product where  
quantity>0").uniqueResult();
```

```
List<Object[]> group =
```

```
session.createQuery("select description,count(*) from Product group by  
description").list();
```

```
Double min =  
(Double) session.createQuery("select min(price) from  
Product").uniqueResult();
```

```
Double max =  
(Double) session.createQuery("select max(price) from  
Product").uniqueResult();
```

```
System.out.println("Total Products: " + total);
```

```
System.out.println("Quantity > 0: " + available);
```

```
group.forEach(o ->
```

```
System.out.println(o[0] + " " + o[1]));
```

```
System.out.println("Min Price: " + min);
```

```
System.out.println("Max Price: " + max);
```

```
break;
```

```
case 7:
```

```
List<Object[]> grp =
```

```
session.createQuery("select description,count(*) from Product group by  
description").list();
```

```
grp.forEach(o ->
```

```
System.out.println(o[0] + " " + o[1]));
```

break;

case 8:

System.**out**.print("Min price: ");

double minp = sc.nextDouble();

System.**out**.print("Max price: ");

double maxp = sc.nextDouble();

Query range = session.createQuery("from Product where price between

:a and :b"); range.setParameter("a", minp);

range.setParameter("b", maxp);

range.list().forEach(System.**out**::println);

break;

case 9:

System.**out**.println("1 Starts With");

System.**out**.println("2 Ends With");

System.**out**.println("3 Contains");

System.**out**.println("4 Exact Length");

int like = sc.nextInt();

if (like == 1) {

System.**out**.print("Enter letters: "); String s = sc.next() +

"%"; session.createQuery("from Product where name

like :x")

.setParameter("x", s)

.list().forEach(System.**out**::println);

}

```

    if (like == 2) {
        System.out.print("Enter letters: "); String s = "%" +
            sc.next(); session.createQuery("from Product where
name like :x")
.setParameter("x", s)
.list().forEach(System.out::println);
    }

    if (like == 3) {
        System.out.print("Enter letters: "); String s = "%" +
            sc.next() + "%"; session.createQuery("from Product
where name like :x")
.setParameter("x", s)
.list().forEach(System.out::println);
    }

    if (like == 4) {
        System.out.print("Enter length: "); int len = sc.nextInt();
        session.createQuery("from Product where
length(name)=:l")
.setParameter("l", len)
.list().forEach(System.out::println);
    }
    break;

case 10:
    System.out.println("Program Ended"); break;
}

```

```
        session.close();  
        sf.close();  
    }  
}
```

Product.java:

```
package com.klu.model;
```

```
import javax.persistence.*;
```

```
@Entity
```

```
@Table(name = "product")
```

```
public class Product {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private int id;
```

```
    private String name;
```

```
    private String description;
```

```
    private double price;
```

```
    private int quantity;
```

```
    public int getId() {
```

```
        return id;
```

```
    }
```

```
public String getName() { return  
    name;  
}
```

```
public String getDescription() { return  
    description;  
}
```

```
public double getPrice()  
    { return price;  
}
```

```
public int getQuantity() {  
    return quantity;  
}
```

```
public void setName(String name) { this.name  
    = name;  
}
```

```
public void setDescription(String description) { this.description  
    = description;  
}
```

```
public void setPrice(double price) { this.price  
    = price;  
}
```

```
    public void setQuantity(int quantity) { this.quantity  
        = quantity;  
    }  
}
```

Hibernate.cfg.xml:

```
package com.klu.model;
```

```
import javax.persistence.*;
```

```
@Entity
```

```
@Table(name = "product")
```

```
public class Product {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    private int id;
```

```
    private String name;
```

```
    private String description;
```

```
    private double price;
```

```
    private int quantity;
```

```
    public int getId() {
```

```
        return id;
```

```
    }
```

```
public String getName() { return  
    name;  
}
```

```
public String getDescription() { return  
    description;  
}
```

```
public double getPrice()  
    { return price;  
}
```

```
public int getQuantity() {  
    return quantity;  
}
```

```
public void setName(String name) { this.name  
    = name;  
}
```

```
public void setDescription(String description) { this.description  
    = description;  
}
```

```
public void setPrice(double price) { this.price  
    = price;  
}
```

```
public void setQuantity(int quantity) { this.quantity  
    = quantity;  
}  
}
```

Pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
        http://maven.apache.org/xsd/maven-4.0.0.xsd">  
    <modelVersion>4.0.0</modelVersion>  
  
    <groupId>com.klu</groupId>  
    <artifactId>HibernateSkill3</artifactId>  
    <version>1.0</version>  
  
    <properties>  
        <maven.compiler.source>1.8</maven.compiler.source>  
        <maven.compiler.target>1.8</maven.compiler.target>  
    </properties>  
  
    <dependencies>  
  
        <dependency>  
            <groupId>org.hibernate</groupId>  
            <artifactId>hibernate-core</artifactId>  
            <version>5.6.15.Final</version>
```


</dependency>

<dependency>

<groupId>javax.persistence</groupId>

<artifactId>javax.persistence-api</artifactId>

<version>2.2</version>

</dependency>

<dependency>

<groupId>mysql</groupId>

<artifactId>mysql-connector-java</artifactId>

<version>8.0.33</version>

</dependency>

<dependency>

<groupId>org.slf4j</groupId>

<artifactId>slf4j-simple</artifactId>

<version>1.7.36</version>

</dependency>

</dependencies>

</project>